$= \frac{d}{dx}\left[50 + 4x + 3\sqrt{x}\right] = 4 + \frac{3}{2\sqrt{x}}$

When $x = 9$, $\frac{dC}{dx} = 4 + \frac{3}{2\sqrt{9}} = 4\frac{1}{2}$ (or) ₹ 4.50

$\therefore$ MC is ₹ 4.50 , when the level of output is 9 units.

Example 6.3

Find the equilibrium price and equilibrium quantity for the following demand and supply functions.

Demand : $x = \frac{1}{2}(5 - p)$ and Supply : $x = 2p$–3.

Solution:

At equilibrium, demand = supply

$\Rightarrow \quad \frac{1}{2}(5 - p) = 2p$–3

5–$p = 4p - 6$

$\Rightarrow p = \frac{11}{5}$

$\therefore$ Equilibrium price: $p_E =$ ₹ $\frac{11}{5}$

Now, put $p = \frac{11}{5}$ in $x = 2p - 3$

We get, $x = 2\left(\frac{11}{5}\right) - 3 = \frac{7}{5}$

$\therefore$ Equilibrium quantity: $x_E = \frac{7}{5}$ units.

Example 6.4

For the demand function $x = \frac{20}{p+1}$, p > 0, find the elasticity of demand with respect to price at a point p = 3. Examine whether the demand is elastic at p = 3.

Solution:

$x = \frac{20}{p+1}$

$\frac{dx}{dp} = \frac{-20}{(p+1)^2}$

Elasticity of demand: $\eta_d = -\frac{p}{x} \cdot \frac{dx}{dp}$

$= -\frac{p}{\left(\frac{20}{(p+1)}\right)} \cdot \frac{-20}{(p+1)^2} = \frac{p}{p+1}$

When p=3, $\eta_d = \frac{3}{4}$ (or) 0.75

Here $|\eta_d| < 1$

$\therefore$ demand is inelastic.

Example 6.5

Find the elasticity of supply for the supply function $x = 2p^2 - 5p + 1$, p> 3.

Solution:

$x = 2p^2 - 5p + 1$

$\frac{dx}{dp} = 4p - 5$

Elasticity of supply: $\eta_s = \frac{p}{x} \cdot \frac{dx}{dp}$

$= \frac{p}{2p^2 - 5p + 1} \cdot (4p - 5) = \frac{4p^2 - 5p}{2p^2 - 5p + 1}$

Example 6.6

If $y = \frac{2x+1}{3x+2}$, then obtain the elasticity at $x = 1$.

Solution:

$y = \frac{2x+1}{3x+2}$

$\frac{dy}{dx} = \frac{(3x+2)(2) - (2x+1)(3)}{(3x+2)^2} = \frac{1}{(3x+2)^2}$

Elasticity: $\eta = \frac{x}{y} \cdot \frac{dy}{dx} = \frac{x}{\left(\frac{2x+1}{3x+2}\right)} \cdot \frac{1}{(3x+2)^2}$

$= \frac{x}{(2x+1)(3x+2)}$

When $x = 1$, $\eta = \frac{1}{15}$

Example 6.7

A demand function is given by $xp^n = k$ where n and k are constants. Prove that elasticity of demand is always constant.

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